> **Note:** Currently, we are doing basic offline handling within our PWA. To improve the experience however, we can add the fetched content into another *DATA_CACHE* and serve that instead of our *offline.html*. This is beyond the scope of this article, though readers are encouraged to check out relevant resources.

To complete the coding for service workers, we now need to add boilerplate code into *index.html* so that service workers are registered, installed and activated by the browser, as shown below:

```
<script>
  // CODELAB: Register service
worker.
  if ("serviceWorker" in navigator)
{
    window.addEventListener("load",
() => {
      navigator.serviceWorker.
register("/service-worker.js").then(reg
=> {
        console.log("Service worker
registered.", reg);
      });
    });
  }
</script>
```

More than installable: Putting in place a custom install experience

Strictly speaking, this task is not a PWA audit error, but more an expected industry standard experience where the PWA audit already shows the Web app as installable. To complete this we add some more boilerplate JavaScript code in *install.js* as shown in the code explanations below, and reference the same from *index.html* with the *<script>* tags the way we already did for *service-worker.js*.

Create an *Install* button, register for a click, listen for *beforeinstallprompt* and also for the app installed events:

```
let deferredInstallPrompt = null;
const installButton = document.
getElementById("butInstall");
installButton.addEventListener("click",
installPWA);

// CODELAB: Add event listener for
beforeinstallprompt event
window.addEventListener("beforeinstallp
rompt", saveBeforeInstallPromptEvent);

// CODELAB: Add event listener for
appinstalled event
window.addEventListener("appinstalled",
logAppInstalled);
```

Save the *beforeinstall* event and show the *Install* button:

```
/**
 * Event handler for
beforeinstallprompt event.
 *   Saves the event & shows install
button.
 *
 * @param {Event} evt
 */
function saveBeforeInstallPromptEvent(
evt) {
  // CODELAB: Add code to save event &
show the install button.
  deferredInstallPrompt = evt;
  installButton.
removeAttribute("hidden");
}
```

Show the *install* prompt and hide the *Install* button:

```
/**
 * Event handler for butInstall - Does
the PWA installation.
 *
 * @param {Event} evt
 */
function installPWA(evt) {
  // CODELAB: Add code show install
prompt & hide the install button.
  // CODELAB: Log user response to
prompt.
  deferredInstallPrompt.prompt();
  // Hide the install button, it can't
be called twice.
  evt.srcElement.setAttribute("hidden",
true);
```

Log the selection and reset variables:

```
  // CODELAB: Log user response to
prompt.
  deferredInstallPrompt.userChoice.
then(choice => {
    if (choice.outcome === "accepted")
{
      console.log("User accepted the
A2HS prompt", choice);
    } else {
      console.log("User dismissed the
A2HS prompt", choice);
    }
    deferredInstallPrompt = null;
  });
}
```

Make a log that our Web app was installed:

```
/**
 * Event handler for appinstalled
event.
 *   Log the installation to analytics
or save the event somehow.
 *
 * @param {Event} evt
 */
function logAppInstalled(evt) {
  // CODELAB: Add code to log the event
  console.log("CoronaVirus Stats Dump
App was installed.", evt);
}
```

Index.html changes for *theme-colour*, *apple-touch-icon* and *Install.js* are all shown in the code segment below:

```
  <meta name="viewport"
content="width=device-width, initial-
scale=1" />
  <meta name="theme-color"
content="#3367D6" />
  <link rel="apple-touch-icon"
```